

Smart Energy Meter - Project Overview

Portable ESP32 smart plug for energy monitoring and mobile control.

1. Project Purpose

This project monitors electrical usage of a connected load and shows voltage, current, power, energy, and cost on an OLED display and a Flutter mobile app. The smart plug can also switch the load ON/OFF using a relay.

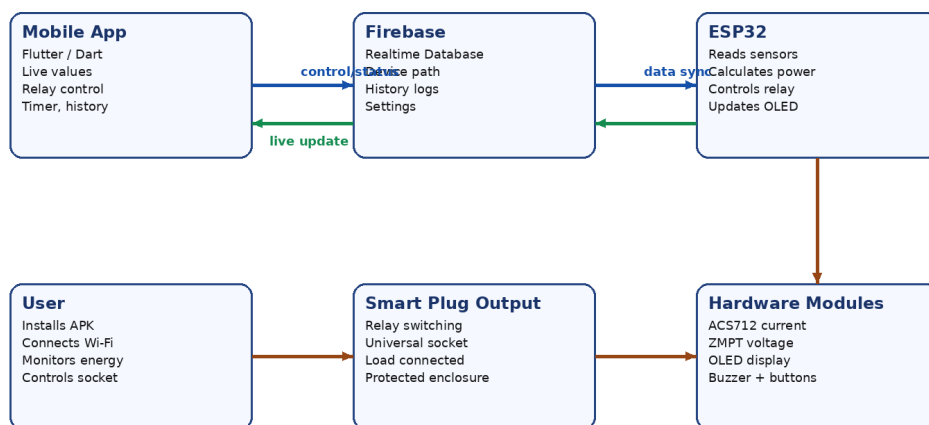
2. Main Features

- Real-time voltage, current, power, energy, and cost monitoring
- OLED display for live values
- Flutter mobile app developed using Dart
- Firebase Realtime Database connection
- Relay ON/OFF control from app and physical button
- Wi-Fi setup page for entering new Wi-Fi SSID and password
- Timer based auto OFF function
- Usage history and analysis screens
- Power-cut handling and status display

3. System Architecture

Smart Energy Meter - System Architecture

Flutter app + Firebase + ESP32 smart plug hardware



Safety note: AC mains side must be enclosed, insulated, fused/protected, and checked by a qualified person.
Do not touch live parts while powered. This diagram is system-level documentation.

4. Important Safety Notes

Project Overview

- This project works near AC mains voltage. Build and test it only with qualified supervision.
- Use proper enclosure, insulation, strain relief, fuse/MCB protection, and safe earthing where required.
- Do not touch any AC/live part while powered.
- This documentation shows system-level connections. Final mains wiring must be checked by a qualified person before use.